

# Python Case Study: Strategic Insights

E-Commerce Data Analysis & Business Intelligence (2021-2022)

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# Project Context & Methodology

Harnessing Python for data-driven decision making in warehouse and marketing ecosystems.



# E-Commerce Strategic Challenges

## Business Background

In a competitive e-commerce landscape, rapid response to divisional needs is critical. This project focuses on solving core challenges for Marketing, Warehouse, and Campaign teams.

**Primary Goal:** Analyze 2021-2022 transaction datasets to optimize stock, evaluate marketing effectiveness, and drive customer loyalty through data-backed rewards.





# Python Environment & Framework

-  **Pandas & Numpy:** Data wrangling and complex numerical computation.
-  **Matplotlib & Seaborn:** High-fidelity statistical visualization and plotting.
-  **Pandas BDay:** Specialized offset logic for business day calculations.
-  **SQLite3:** In-memory SQL engine for efficient relational data merging.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

*Standardized library ingestion for consistent analysis pipelines.*



# Integrated Dataset Architecture



## Order Detail

Transaction IDs, SKUs, pricing (before/after discount), and validation flags (is\_valid).



## SKU Detail

Product catalog containing SKU names, COGS, and broad product categories.



## Customer Detail

Anonymized user IDs and historical registration dates for cohort analysis.

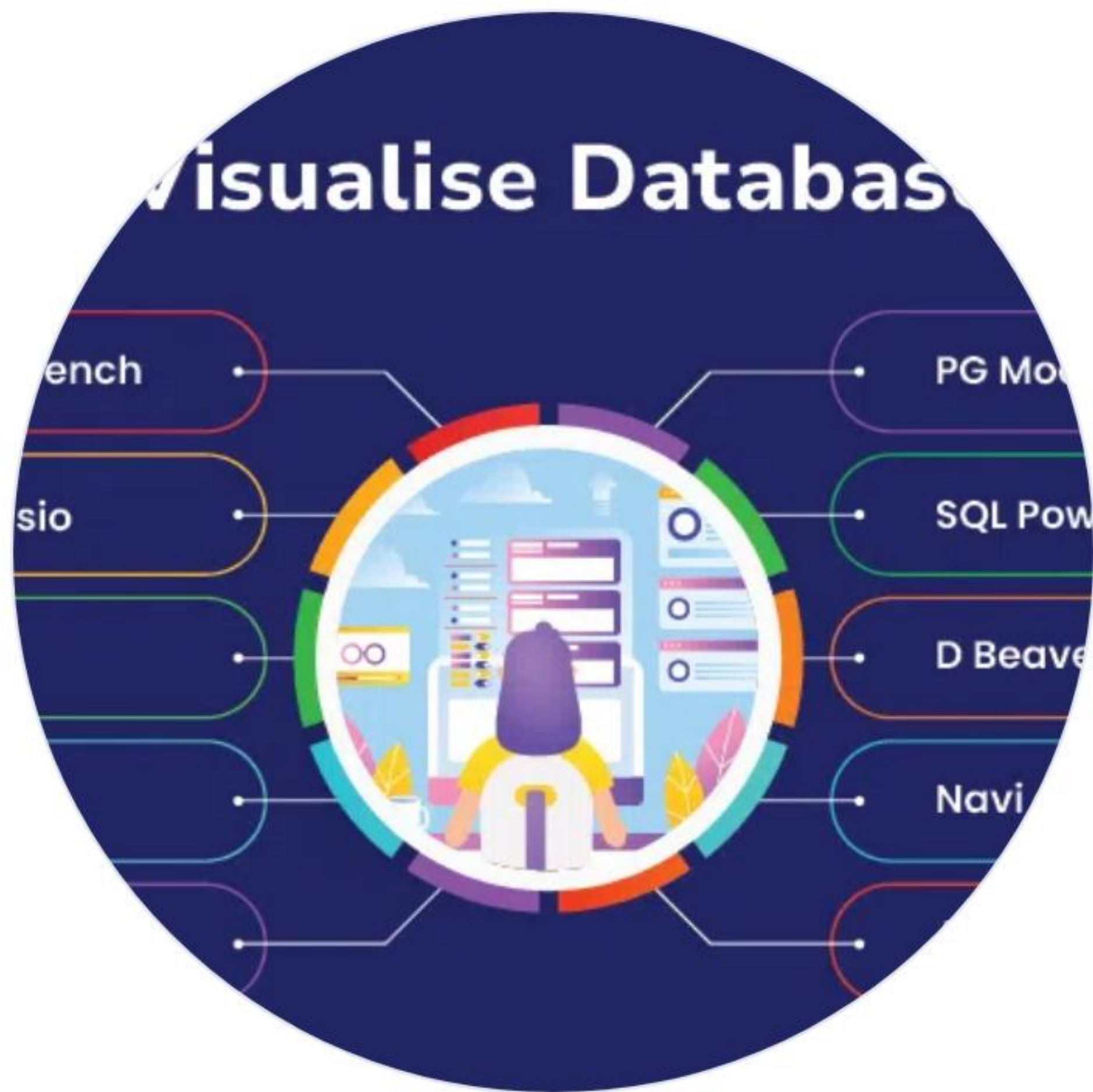


## Payment Detail

Lookup table for transaction payment methods and financial instruments.



# SQL Relational Integration Logic



## Memory-Based Merging

Utilizing a `SQLite :memory:` connection allowed for high-speed relational joins across distributed CSV sources.

-  **Left Join** on Payment ID
-  **Left Join** on SKU ID
-  **Left Join** on Customer ID

Result: A unified master dataframe containing full transaction, product, and user context.



# Data Transformation & Sanitization

| Feature Column  | Source Type | Target Format | Rational / Purpose                          |
|-----------------|-------------|---------------|---------------------------------------------|
| before_discount | float64     | Integer       | Memory efficiency for currency values.      |
| after_discount  | float64     | Integer       | Streamlined reporting for financial totals. |
| order_date      | object      | Datetime64    | Enabled year-over-year temporal analysis.   |
| registered_date | object      | Datetime64    | Enabled customer onboarding speed metrics.  |

*Standardization ensures data integrity during grouping and aggregation operations.*

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MARKETING CASE STUDY

# Festival Akhir Tahun

Predicting gift allocation based on high-performance sales metrics.



# Problem Statement & Logic

"We need to predict gift estimates for the Festival Akhir Tahun competition. Determine the **Top 5 Products** from the **Mobiles & Tablets** category during **2022** based on total sales quantity (valid = 1)."

## Strategic Filters

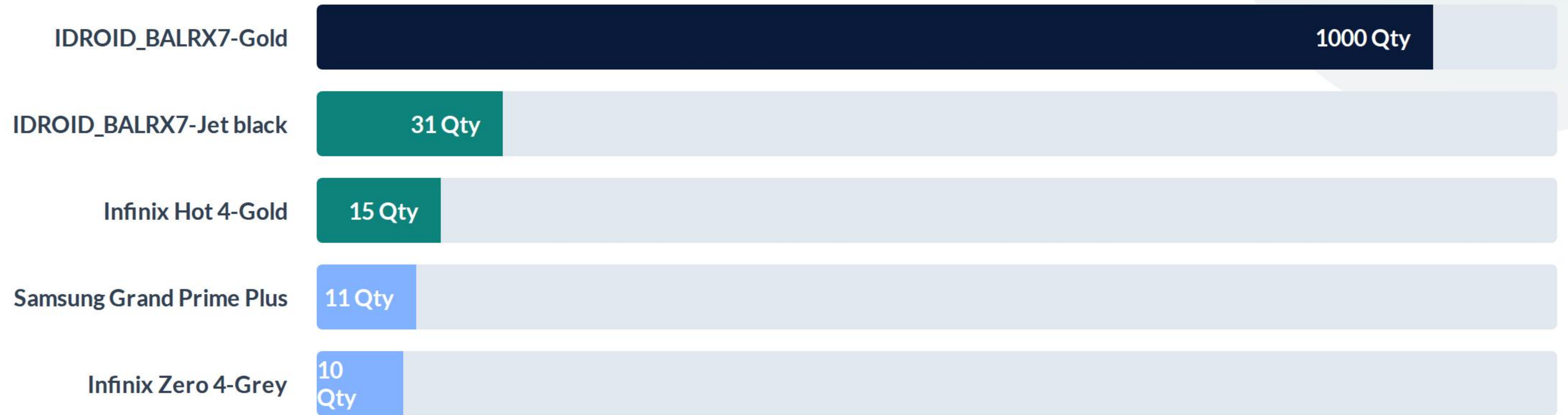
- 🔹 Year == 2022
- ✓ Is\_Valid == 1
- 📱 Category == 'Mobiles & Tablets'

```
# Filter & Aggregation
filter_data = df[
    (df['order_date'].dt.year == 2022) &
    (df['is_valid'] == 1) &
    (df['category'] == 'Mobiles & Tablets')
]

top5 = filter_data.groupby('sku_name')\
    .agg(total_qty=('qty_ordered', 'sum'))\
    .sort_values('total_qty', ascending=False)\
    .head(5)
```



# Market Performance Results (2022)



**Data Insight:** IDROID\_BALRX7-Gold shows an extreme sales spike, suggesting a successful mass-marketing campaign or exclusive B2B bundle in 2022.



# Strategic Action Plan

## Market Dominance

IDROID\_BALRX7-Gold accounts for nearly 95% of the Top 5 volume. This product is the safest choice for campaign prizes due to proven market acceptance.

## Diverse Reward Mix

Use Infinix and Samsung models as secondary tier prizes to provide variety without over-saturating the reward perception.





# Questions?

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# Image Sources



Thumbnail for <https://thefintechtimes.com/wp-content/uploads/2023/06/ecommerce.jpg>

Source: [thefintechtimes.com](https://thefintechtimes.com)

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<https://media.geeksforgeeks.org/wp-content/uploads/20240703171512/Tools-to-Visualise-Database-Schema.webp>

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